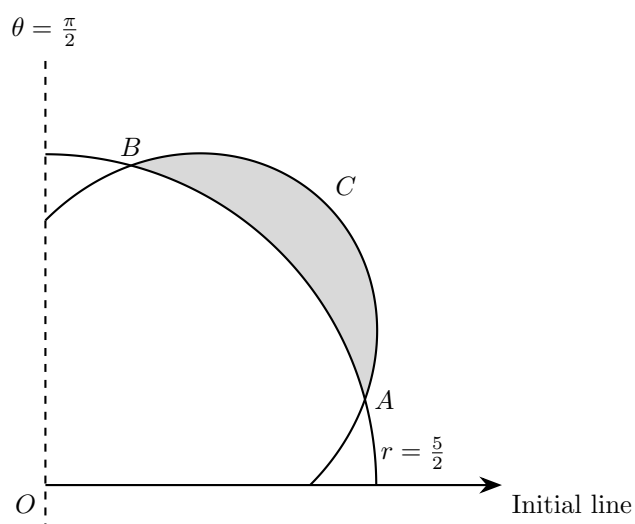


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1. The diagram shows part of the curve C with polar equation

$$r = 2 + \sin 2\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

and the circle

$$r = \frac{5}{2}$$

The curve and the circle intersect at the points A and B , as shown.

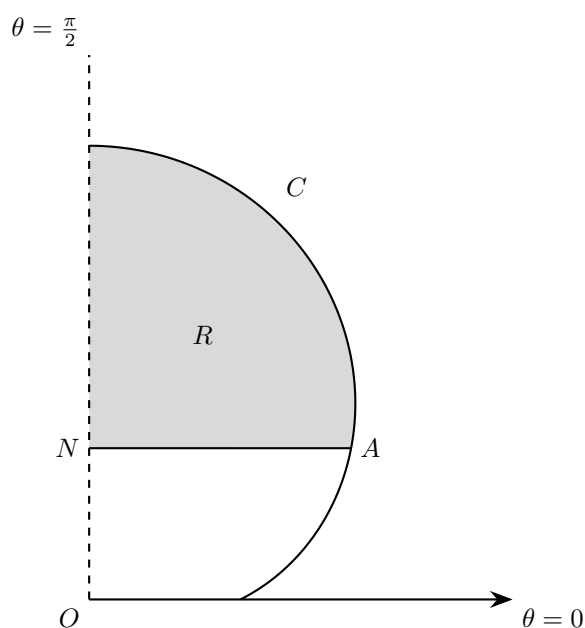
Find, correct to three significant figures, the area of the shaded region enclosed by C and the circle.

[9]

2. A curve has polar equation $r = a \sin^2 3\theta$, where $a > 0$ and $0 \leq \theta < 2\pi$.

(a) Sketch the curve. [2]

(b) Find, in exact form, the total area of the regions enclosed by the curve. [7]



3. The curve C shown in the diagram above has polar equation

$$r = 1 + 2\sin\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

The point A on C has y -coordinate 1.

The point N lies on the line $\theta = \frac{\pi}{2}$ and AN is perpendicular to the line $\theta = \frac{\pi}{2}$.

The finite region R , shown shaded in the diagram above, is bounded by the curve C , the line $\theta = \frac{\pi}{2}$ and the line AN .

Find the exact area of the shaded region R .

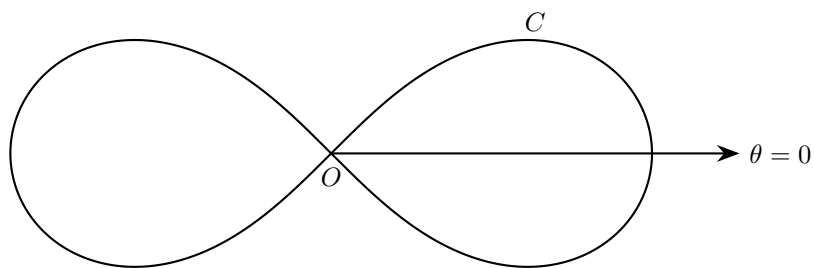
[9]

4. The curve C has polar equation

$$r^2 = \frac{\pi^2 - (\theta - \pi)^2}{1 + (\theta - \pi)^2} \quad 0 \leq \theta \leq 2\pi$$

(a) Sketch C and state the polar coordinates of the point on C furthest from the pole. [4]

(b) Using the substitution $u = \theta - \pi$, or otherwise, find the exact area enclosed by C . [6]



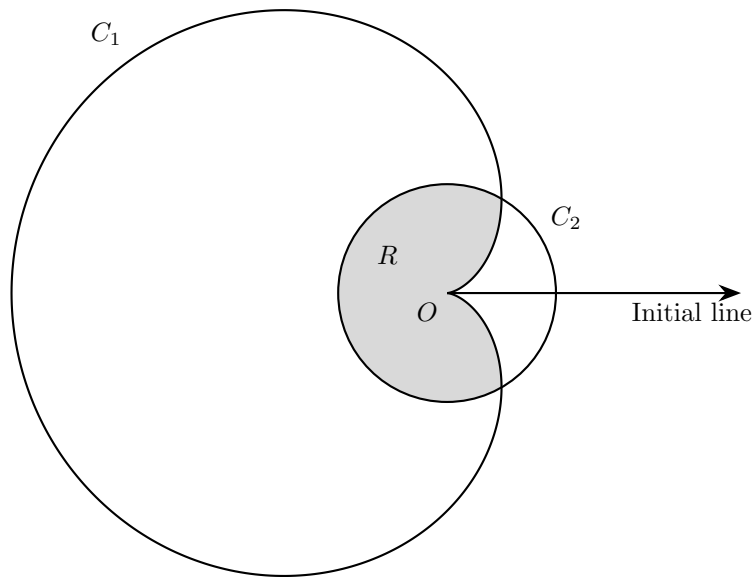
5. The diagram above shows the curve C with polar equation $r^2 = 8 \cos 2\theta$.

(a) Show that a cartesian equation of C is

$$(x^2 + y^2)^2 = 8(x^2 - y^2) \quad [4]$$

(b) Show that the coordinate axes are lines of symmetry of C . [2]

(c) Find the exact area enclosed by each loop of C . [5]



6. The diagram above shows the curve C_1 with polar equation $r = 2a(1 - \cos \theta)$, $0 \leq \theta \leq 2\pi$, and the circle C_2 with polar equation $r = a$, $0 \leq \theta \leq 2\pi$, where a is a positive constant.

- (a) Find, in terms of a , the polar coordinates of the points where C_1 meets C_2 . [3]

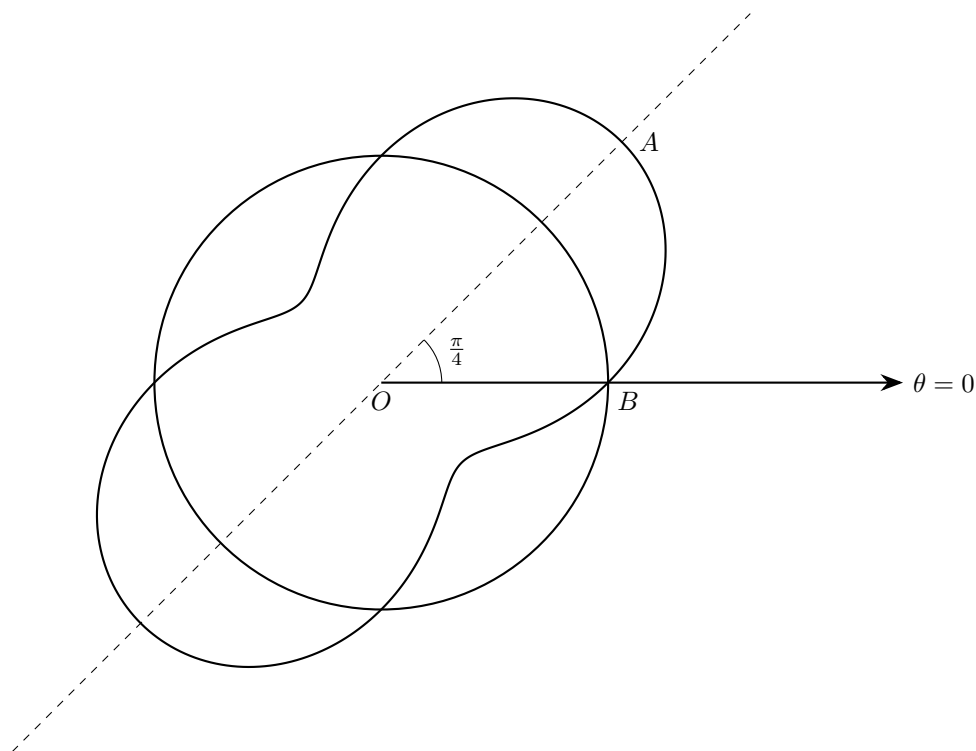
The common region enclosed by C_1 and C_2 is shaded in the diagram above.

- (b) Find the area of the shaded region R , giving your answer in the form

$$\frac{1}{12}a^2 (p\pi + q\sqrt{3})$$

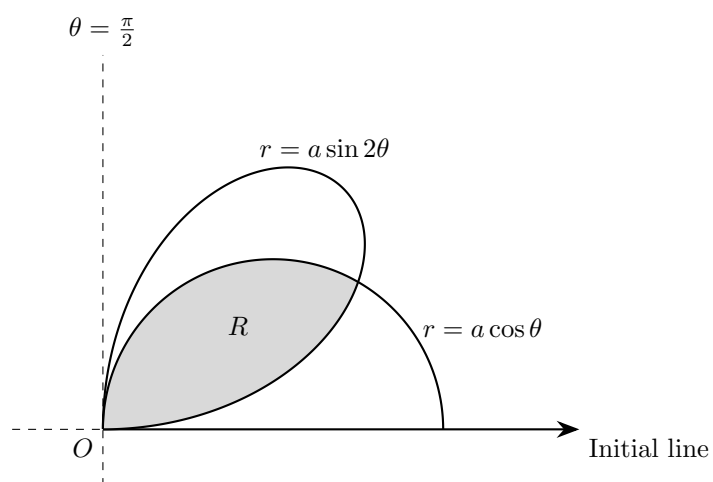
where p and q are integers to be found.

[7]



7. The diagram above shows the curve C with polar equation $r = a(2 + \sin 2\theta)$ and the circle $r = 2a$ for $-\pi \leq \theta \leq \pi$, where a is a positive constant.

- (a) Write down the polar coordinates of the points A and B . [2]
- (b) Explain why C is symmetrical about the line $\theta = \frac{\pi}{4}$. [2]
- (c) Find, in terms of a , the exact total area of the regions that lie inside C and outside the circle. [5]



8. In the first quadrant, the curve $r = a \sin 2\theta$ intersects the circle $r = a \cos \theta$ at the pole and at one other point.

The region R shown consists of the points that lie inside both curves.

Find the exact area of R in terms of a .

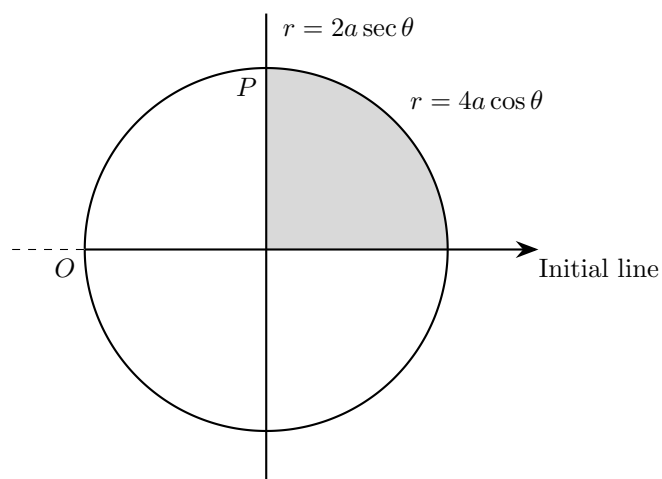
[8]

9. (a) On the same diagram, sketch the curve C_1 with polar equation

$$r = 2(1 + \cos \theta), \quad 0 \leq \theta \leq 2\pi$$

and the curve C_2 with polar equation $\theta = \frac{2\pi}{3}$. [3]

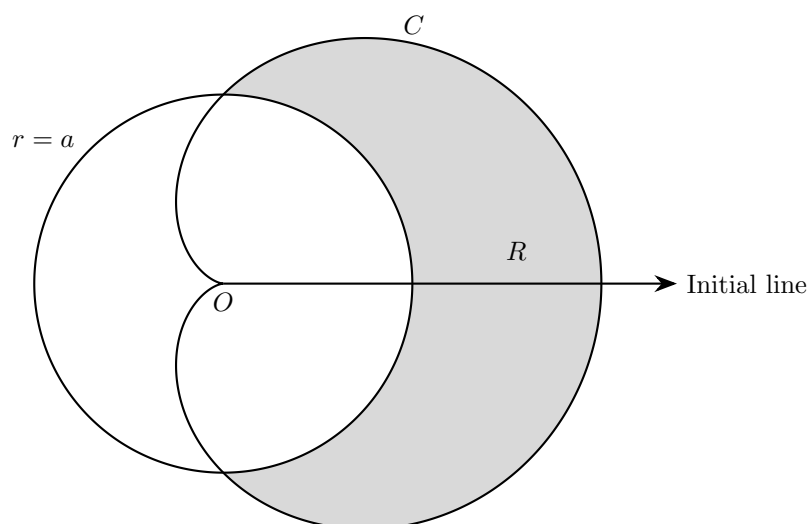
- (b) Find the area of the smaller region bounded by C_1 and C_2 . [6]



10. The diagram shows a sketch of the curves with polar equations

$$r = 4a \cos \theta \text{ and } r = 2a \sec \theta, \quad a > 0$$

- (a) Find the polar coordinates of the point of intersection P of the two curves in the first quadrant. [4]
- (b) Find the area, shaded in the figure, bounded by the two curves and by the initial line $\theta = 0$, giving your answer in terms of a and π . [6]



11. The figure above shows a sketch of the curve C with polar equation

$$r = a(1 + \cos \theta)$$

and the circle

$$r = a$$

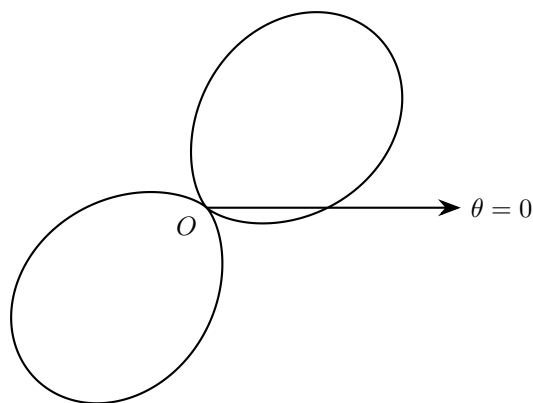
where $a > 0$ and $-\pi \leq \theta \leq \pi$.

The shaded region inside C and outside the circle has area

$$18 + \frac{9\pi}{4}$$

Find the value of a .

[8]



12. The diagram above shows the curve with polar equation $r = 1 + \sin 2\theta$ for $0 \leq \theta \leq 2\pi$.

- (a) Find the values of θ at the pole. [1]
- (b) Find the polar coordinates of the points on the curve where r takes its maximum value. [2]
- (c) Find the exact total area enclosed by the curve. [4]
- (d) Find a Cartesian equation for the curve. [2]